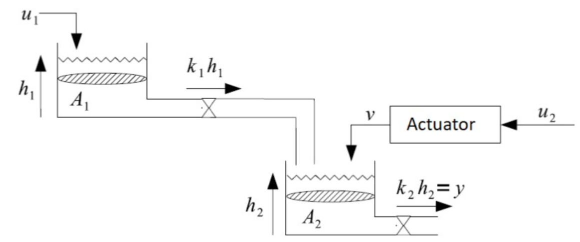
Consider the following system



Where

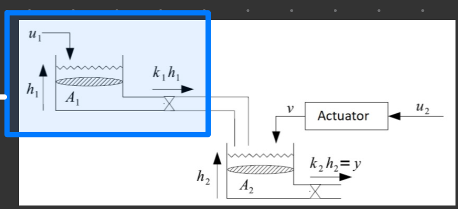
- $u_1(t)$ is a volume flow $\left[\frac{m^3}{min}\right]$
- $h_1(t), h_2(t)$ are the liquid levels in the tanks
- The tank area $A_1 = A_2 = 2\ m^2$
- output flows $q_1(t), q_2(t)$ are governed by $q_i = k_i h_i(t)$
- The actuator is model by the following equations

$$\dot{x_1}(t) = \frac{1}{h_1 + 1} x_1(t)$$

$$\dot{x_2}(t) = -x_1(t) - x_2(t) + u_2(t)$$

$$v(t) = x_2(t)$$

Considering that we are interested in the output flow $q_2(t)$, determine the system equation.



TANK 1

$$\dot{x}_1(t) = \frac{1}{S}\left(w_1 - K_i x_1(t)\right)$$

$$\dot{x}_2(t) = \frac{1}{S}\left(K_i x_1(t) + V(t) - K_2 x_2(t)\right)$$

$$y(t) = K_2 x_2$$

$$\dot{x}_1(t) = x_1^2(t)\left(x_2(t) - 2\right) + w(t)$$

$$\dot{x}_2(t) = \left(x_1(t) - \alpha\right)\left(x_2(t) - 2\right) - x_2(t)$$

$$y(t) = x_1(t) x_2(t)$$

EQUILIBRIUM $\left((\bar{x}_1, \bar{x}_2), \bar{w}\right)$

$w(t) = 0$

$$\begin{cases} 0 = x_1^2(t)\left(x_2(t) - 2\right) \\ 0 = \left(x_1(t) - \alpha\right)\left(x_2(t) - 2\right) - x_2(t) \end{cases} \longrightarrow \text{TWO CASES: } x_1(t) = 0 \begin{cases} 0 = 0 \\ 0 = -\alpha\left(x_2(t) - 2\right) - x_2(t); \end{cases}$$

$y(t) = x_1(t) x_2(t)$

$y(t) = 0$

$x_2(t) + \alpha x_2(t) = 2\alpha$

$x_2(t) = \frac{2\alpha}{1-\alpha}$

$\rightarrow E_1: \left(\left(0; \frac{2\alpha}{1-\alpha}\right), 0\right)$

$$x_2(t) = 2 \begin{cases} 0 = 0 \\ 0 = -2 \ \text{UNSAT} \end{cases}$$

LINEARIZED SYSTEM

$$\delta w(t) = w(t) - \bar{w} = w(t)$$

$$\delta x_1(t) = x_1(t) - \bar{x}_1 = x_1(t)$$

$$\delta x_2(t) = x_2(t) - \bar{x}_2 = x_2(t) - \frac{2\alpha}{1-\alpha}$$

$$\delta y(t) = y(t) - \bar{y} = y(t)$$

$\bar{x}_1(t) \bar{x}_2(t) = 0$

STABILITY

$$\begin{cases} \dot{x}_1(t) = x_1^2(t)\left(x_2(t) - 2\right) + w(t) \\ \dot{x}_2(t) = \left(x_1(t) - \alpha\right)\left(x_2(t) - 2\right) - x_2(t) \\ y(t) = x_1(t)x_2(t) \end{cases}$$

$$Eq: \left(\left(0; \frac{2\alpha}{1-\alpha}\right); 0\right)$$

$$A = \begin{bmatrix} 2x_1(x_2(t)-2) & x_1^2(t) \\ x_2(t)-2 & x_1(t)-\alpha-1 \end{bmatrix}\Bigg|_{\bar{x},\bar{w}} = \begin{bmatrix} 0 & 0 \\ \frac{2\alpha}{1-\alpha}-2 & -\alpha-1 \end{bmatrix}$$

$$\begin{cases} \delta\dot{x}(t) = A\delta x(t) + B\delta w(t) \\ \delta y(t) = C\delta x(t) + D\delta w(t) \end{cases}$$

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} x_2(t) & x_1(t) \end{bmatrix}\Big|_{\bar{x},\bar{w}} = \begin{bmatrix} \frac{2\alpha}{1-\alpha} ; & 0 \end{bmatrix}$$

$$D = 0$$

$$\begin{cases} \delta\dot{x}_1(t) = 0x_1(t) + 0x_2(t) + 1w(t) = w(t) \\ \delta\dot{x}_2(t) = \left(\frac{2\alpha}{1-\alpha} - 2\right)x_1(t) + (-\alpha-1)x_2(t) + 0w(t) \\ \delta y(t) = \frac{2\alpha}{1-\alpha}x_1(t) + 0x_2(t) + 0w(t) \end{cases}$$

$$DET(\lambda I - A) = DET\left(\begin{bmatrix} \lambda & 0 \\ -\frac{2\alpha}{1-\alpha}+2 & \lambda+\alpha+1 \end{bmatrix}\right)$$

$$\Rightarrow \lambda(\lambda+\alpha+1) - 0 = 0 \rightarrow$$

- $\lambda = 0$
- $\lambda = -\alpha-1$

$-\alpha-1 < 0$
$\alpha > -1$
SIMPLY STABLE

$-\alpha-1 > 0$
$\alpha < -1$
UNSTABLE